Quantitative Residual-Risk Validation for SOTIF

From acceptance criteria to scenario-based evidence for an L4 automated driving system

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Abstract

This report collects, in one place, the quantitative chain needed to argue that the residual risk of an automated driving system is acceptable under ISO 21448 (SOTIF). It starts from an acceptance criterion, derives validation targets for field operational testing, and shows why those targets become infeasible for an L4 system. It then develops scenario-based residual-risk estimation (exposure, parameter distributions with a Gaussian copula, importance sampling), works through a fully synthetic example with 100 logical and 1,000 concrete scenarios for unintended braking, and derives the field data, statistics and safety performance indicators (SPIs) that the argument needs. It closes with a vehicle-level validation strategy for L4, the role of a scenario database, automatic scenario identification, the granularity of ODD tags, arguments for completeness, and how the approach scales from ACC to an urban robotaxi. Every statement taken from the literature carries a pinpoint citation (section, equation, table, page or slide), so the reader can check it at the source.

Highlights

- The Annex C validation target $\tau = -\ln(1-C)/R_{\mathrm{HB}}$ rests on five assumptions that an assessor can attack; each is stated with its source.
- Counting critical scenario runs is not a risk estimate; only weights $w = f/q$ connect test runs to real-world exposure.
- In the synthetic example, the Bayesian 95% bound passes the budget while a Bonferroni bound over 120 estimates fails: the inference method decides the evidence plan.
- Ignoring parameter dependence changed $P(\text{collision} \mid \text{HB})$ by a factor of 1.06 to 3.2 in the example.
- Completeness of a scenario concept cannot be proven; it is argued from construction, explained time, saturation, a risk-weighted remainder and operational monitoring.

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1 Introduction

1.1 Purpose and scope

The report supports a SOTIF engineer who must build and defend the quantitative part of a residual-risk argument for a Level 4 system. It covers the statistics behind validation targets, scenario-based risk estimation, the data that feeds it, and how the pieces fit into a vehicle-level validation strategy. It does not cover ISO 26262 hardware metrics or cybersecurity.

1.2 How sources are cited

Citations give the location inside the source in square brackets, for example [9, Sec. IV-B, Eq. (1)] or [11, p. 3, Eq. 2]. The locations were checked against the source texts in October 2026. Three conventions apply:

- **Standards.** The ISO standards themselves were not available to the author of this report. Clause, annex and page references to ISO 21448, ISO/TS 5083 and UL 4600 are therefore taken from the secondary source named in the same citation, for example "ISO 21448, Annex C.3, p. 137, as cited in [10, Sec. III-C]". Check them against a licensed copy before relying on them.
- **Location not verified.** Where only the source, not the location, could be checked, the citation says so.
- **Own results.** Numbers computed for this report are marked *(this report, Sec. x / Fig. y)*. All example data are synthetic, marked ^{S}, and describe no real system. The code that produces them is listed in Appendix A.

1.3 Structure

Section 2 summarises the SOTIF framework for quantitative validation. Section 3 derives validation targets from acceptance criteria. Section 4 develops scenario-based risk estimation, and Section 5 the Gaussian copula for scenario parameters. Section 6 works through the unintended-braking example. Section 7 covers field data, statistics and SPIs. Section 8 treats the L4 vehicle-level strategy, the scenario database and completeness. Section 9 discusses scaling to ACC and a robotaxi, and Section 10 lists limitations.

2 The SOTIF framework for quantitative validation

ISO 21448 asks for the absence of unreasonable risk rather than absolute safety, and structures the argument around hazards, triggering conditions and residual risk [12, Sec. 1.1, pp. 2–3]. The clauses relevant here are:

- **Clause 6:** identification and evaluation of hazards; an acceptance criterion of the residual risk is determined for each hazard [12, Sec. 2.1, p. 8].
- **Clause 9:** the verification and validation strategy, including validation targets, "the system's minimum performance level that is to be shown during validation" [12, Sec. 2.1, p. 9].
- **Clauses 10 and 11:** evaluation of known and unknown hazardous scenarios. For unknown scenarios, the standard asks to demonstrate that the residual risk meets the acceptance criteria with sufficient confidence [12, Sec. 2.1, p. 9].
- **Clause 12:** evaluation of whether the SOTIF objectives are achieved, as input to release [12, Sec. 2.1, p. 9].

- **Clause 13:** operation-phase activities, centred on field monitoring [12, Sec. 2.1, pp. 9–10].
- **Annex C:** guidance on deriving validation targets and on validation [12, Sec. 2.2, p. 10].

Acceptance criteria are justified by risk acceptance rationales such as ALARP, MEM, positive risk balance and GAMAB [10, Sec. III-B].

For L4 systems, ISO 21448 is complemented by:

- **ISO/TS 5083**, the ADS-level standard. Betschinske et al. quote its statement that, for L3, "a similar amount of real-world test driving as for state-of-the-art L2 systems" seems appropriate (ISO/TS 5083, p. 146, as cited in [10, Sec. I]), and argue that carrying the L2 derivation to higher levels gives infeasible targets [10, Sec. I].
- **ANSI/UL 4600**, which defines safety performance indicators (Sec. 7.3) (UL 4600 Ch. 16, as cited in [21, Sec. 2] and [22, slide 12]).
- **The UNECE NATM multi-pillar approach** (audit, simulation, track, real-world testing, in-service monitoring; ECE/TRANS/WP.29/2021/61), on which Commission Implementing Regulation (EU) 2022/1426 builds [8, web article, no sections].

3 From acceptance criterion to validation target

3.1 Decomposition of the acceptance criterion

ISO 21448 Annex C.2 factorises the acceptance criterion A_H for a harm H as

$$A_H = R_{\mathrm{HB}} \cdot P(E \mid \mathrm{HB}) \cdot P(\bar{C} \mid E) \cdot P(S \mid \bar{C}) \tag{1}$$

(Annex C.2, as rendered in [9, Sec. IV-B(a), Eq. (1)] and [10, Sec. V, Eqs. (6)–(7)]). Here R_{HB} is the rate of the hazardous behaviour, $P(E \mid \mathrm{HB})$ the probability of the exposing scenario, $P(\bar{C} \mid E)$ the probability of non-controllability and $P(S \mid \bar{C})$ the probability of the severity. Solving for R_{HB} gives the tolerable hazardous-behaviour rate.

Putze et al. raise three objections [9, Sec. IV-B(b)]:

1. A correct chain-rule decomposition needs further conditioning, e.g. $P(S \mid \bar{C}, E, \mathrm{HB})$ instead of $P(S \mid \bar{C})$.
2. Annex C.2.1 omits the aggregation over several hazardous behaviours leading to the same harm, without flagging this as a simplification.
3. Field-data quantities are not system-independent: exposure depends on route planning and the ODD; their example is the probability of a truck in front [9, Sec. IV-B(c)].

They therefore bound the harm probability over triples of triggering condition T, behaviour B and event E:

$$P(H) \leq \sum_{T,B,E} P(T)\, P(B \mid T)\, P(E \mid B, T)\, P(H \mid E, B, T) \tag{2}$$

[9, Sec. IV-B(b), Eq. (2); severity-stratified form Eq. (3)], and conclude that "it is not sufficient to derive a single validation target" [9, Sec. IV-B(b), last paragraph].

3.2 The zero-failure test

Assuming hazardous behaviours arrive as a Poisson process, the exposure τ needed to show rate $\leq R_{\mathrm{HB}}$ at confidence C with no observed event is

$$\tau = \frac{-\ln(1 - C)}{R_{\mathrm{HB}}} \tag{3}$$

(Annex C, as rendered in [9, Sec. IV-B(a)]; real-world driving requirement of Annex C.3, ISO 21448 p. 137, as cited in [10, Sec. III-C, Eq. (5)]). Betschinske et al. write the test as $H_0 : R > A_H$ against $H_1 : R \le A_H$ with the type-I error bound $P(X \le k) \le \alpha$, $\lambda = A_H \tau_k$ [10, Sec. III-C, Eqs. (3)–(4)].

Notation trap. In Putze et al. α denotes the confidence level [9, Sec. IV-B(a)]; in Betschinske et al. α is the type-I error, so their Eq. (5) reads $-\ln(\alpha)/A_H$ [10, Sec. III-C, Eq. (5)]. This report writes C for confidence throughout.

At $C = 0.95$, $-\ln 0.05 \approx 3.00$ (the "rule of three"). Kalra and Paddock use the per-mile form $n = \ln(1-C)/\ln(R)$ with reliability $R = 1-F$ [11, Sec. 1, p. 3, Eq. 2]. With the US fatality rate of 1.09 per 100 million miles in 2013 [11, Introduction, p. 2], this gives 275 million failure-free miles, or about 12.5 years for 100 vehicles driving 24/7 at 25 mph [11, Sec. 1, p. 3; Table 1, p. 10]. Demonstrating a fatality rate 20% below the human rate with 95% confidence and 80% power takes more than 11 billion miles, about 518 years [11, Sec. 3, p. 8].

Schaller et al. give the example criterion "overlooking a car on the other lane is at most 10^{-9} per hour", the order of magnitude used in civil aviation for extremely improbable failure conditions, and call the testing needed to prove it prohibitive [12, Sec. 5.1, p. 14]. Equation (3) gives 3.0×10^9 h, about 342,000 vehicle-years of 24/7 operation *(this report, Sec. 3)*.

3.3 Plans that allow k events

If the plan allows k events, the pass condition $P(X \le k \mid \lambda) \le 1 - C$ is solved by

$$\tau_k = \frac{\lambda_k(C)}{R_{\mathrm{HB}}}, \qquad \lambda_k(C) = \tfrac{1}{2}\,\chi^2_{2(k+1)}(C), \tag{4}$$

the classical relation between Poisson confidence limits and the χ^2 distribution [24, location not verified]. Table 1 gives the factors *(this report, computed with SciPy)*.

Table 1: Factors $\lambda_k(C)$ for plans allowing k events *(this report, computed)*.

$C \setminus k$	0	1	2	3	4	5
0.90	2.303	3.890	5.322	6.681	7.994	9.275
0.95	2.996	4.744	6.296	7.754	9.154	10.513
0.99	4.605	6.638	8.406	10.045	11.605	13.108

k and τ must be fixed before testing: under classical hypothesis testing, the test cannot be extended after a failure without inflating the type-I error [10, Sec. III-C, item 6]. Raising C from 0.95 to 0.99 multiplies τ by $\ln 0.01/\ln 0.05 = 1.54$; allowing $k = 1$ at $C = 0.95$ multiplies it by 1.58 *(this report, Table 1)*.

3.4 What a passed test silently assumes

1. **Homogeneous Poisson process**: constant rate and independent events (the Poisson assumption in [9, Sec. IV-B(a)]).
2. **Representative exposure**: the tested mix of conditions equals the deployed mix; the factors in Eq. (1) are system-dependent [9, Sec. IV-B(b)–(c)].
3. **Frozen software**: a fix invalidates earlier hours unless carry-over is argued; reusing earlier systems' data is a main validity gap [10, Sec. V, RA "Decomposition of Hazardous Behavior" and "Bayesian Statistics"].

4. **One behaviour per harm**: otherwise rates add, Eq. (2) [9, Sec. IV-B(b)]. Aggregating separately validated terms gives a joint confidence of at least $1-(\alpha_1+\alpha_2)$ (ISO 21448 p. 165, as cited in [10, Sec. V, RA "Decomposition of Hazardous Behavior"]).
5. **Frequentist meaning**: a system exactly at the limit passes with probability at most $1-C$; a posterior statement needs a prior. The flat prior Gamma$(1,0)$ reproduces the base model, Jeffreys' Gamma$(0.5,0)$ lowers the requirement [10, Sec. V, RA "Bayesian Statistics"].

3.5 Ways to reduce the effort

Betschinske et al. review reduction approaches. Sequential testing (Wald's SPRT) allows optional stopping only if both hypotheses are set at or stricter than the acceptance threshold, $\theta_0, \theta_1 \geq \theta_{A_H}$, and gives a reduction only if $\beta > 1-c$ with $c = \frac{\theta_0}{\theta_1}\alpha^{\theta_0/\theta_1}$ [10, Sec. V, RA "Sequential Testing", Eqs. (14)–(15)]. Further approaches are decomposition of the hazardous behaviour [10, Sec. V, Eqs. (8)–(9)], high-yield route profiles with importance-sampling correction [10, Sec. V, RA "High Yield Route Profiles", Fig. 12], extreme value analysis [10, Sec. V, RA "Extreme Value Analysis", Table IX] and scenario-based testing [10, Sec. V, RA "Scenario Based Testing", Table XI]. Direct mileage-based validation of upper bounds on harm probability is not feasible for ADS [9, Sec. IV-A(b), citing Wachenfeld and Winner], and a residual dependence on field testing is likely to remain [10, Sec. VII].

4 Scenario-based residual risk

4.1 Direction of the calculation

A HARA is built from the system outward: hazardous behaviour, hazardous event, scenarios. Probability flows the other way: the world produces situations at some rate, and the system responds with a hazardous behaviour or not. For the risk calculation the scenario is the random variable and the behaviour is conditioned on it. Logical scenarios must partition the ODD; overlapping scenarios double-count exposure, and then only the bound (2) holds [9, Sec. IV-B(b), Eq. (2)].

4.2 The estimator

For a severity class s and logical scenarios ℓ with exposure ν_ℓ (occurrences or time share per operating hour) and parameter distribution $f_\ell(\theta)$:

$$R_s = \sum_\ell \nu_\ell \, p_{\ell,s}, \qquad p_{\ell,s} = \int P(\mathrm{HB} \mid \theta, \ell)\, P(S \geq s \mid \mathrm{HB}, \theta, \ell)\, f_\ell(\theta)\, d\theta. \tag{5}$$

De Gelder computes risk in the same spirit, by combining the outcome of virtual simulations with the estimated likelihood of scenarios, and decomposes it into exposure, crash probability and severity [14, Ch. 1 and Ch. 7, Secs. 7.3.2, 7.3.4, 7.3.5]. From N concrete runs drawn from a test distribution q_ℓ, the importance-sampling estimator is

$$\hat{p}_{\ell,s} = \frac{1}{N}\sum_{i=1}^{N} w_i h_i, \qquad w_i = \frac{f_\ell(\theta_i)}{q_\ell(\theta_i)}, \qquad \mathrm{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} \tag{6}$$

(importance sampling and effective sample size: [31, Ch. 9]). Zhao et al. apply importance sampling with cross-entropy-optimised proposals to lane-change scenarios from naturalistic data and report acceleration factors of about 2,000 to 20,000 [20, Abstract].

4.3 Counting critical runs is not a risk estimate

Figure 1 shows why. In a toy cut-in model, a uniform test design puts 5.4% of its runs in the failure region, while the real-world probability of failure per cut-in is 4.9×10^{-5} *(this report, Fig. 1[S])*. Non-critical runs are evidence too. Falsification and criticality search find triggering conditions (Clause 10), but do not estimate risk unless q is known. Figure 2 shows importance sampling converging long before plain Monte Carlo sees its first collision.

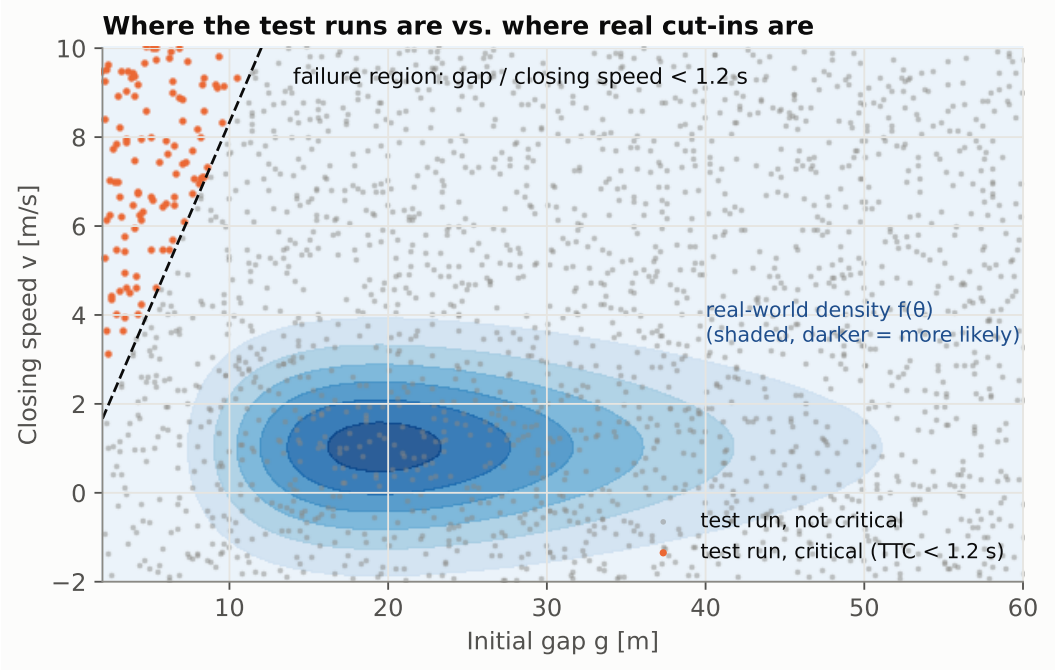


Figure 1: Toy cut-in[S]: test runs of a uniform design against the real-world density. 5.4% of runs are critical; the real-world failure probability is 4.9×10^{-5} *(this report, computed)*.

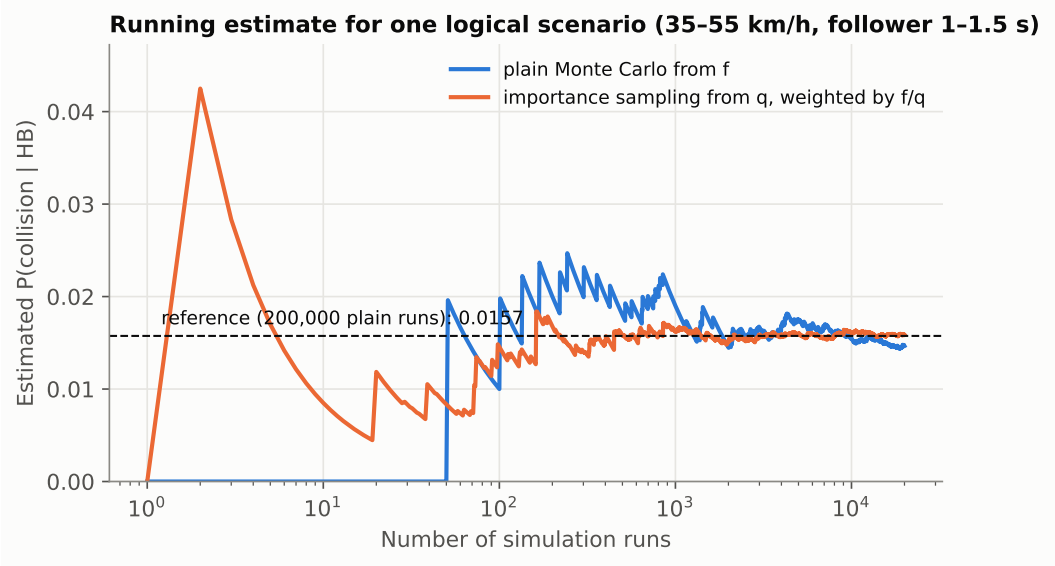


Figure 2: Running estimate of $P(\text{collision} \mid \text{HB})$ for one logical scenario[S]: plain Monte Carlo versus importance sampling, against a 200,000-run reference of 0.0157 *(this report, computed)*.

4.4 Relation to validation targets

The single target τ of Eq. (3) is replaced by one claim with several evidence targets:

$$\sum_{\ell} \nu_{\ell}^{U}\, p_{\ell,s}^{U} + R_{\text{unknown}}^{U} \le A_s \quad \text{at confidence } C, \qquad \sum_{i} (1 - C_i) \le 1 - C, \tag{7}$$

where superscript U marks upper confidence bounds and the confidence budget is split by Bonferroni's inequality.

Table 2: Evidence targets that replace the single field-test target.

Term	Validation target	Measured in
ν_{ℓ}^{U}	enough naturalistic data per scenario class (Poisson bound on counts)	hours of ODD data
f_{ℓ}	estimation error of the fitted distribution below a threshold; de Gelder quantifies completeness via the expected approximation error of a KDE [14, Ch. 3]	scenario instances
p_{ℓ}^{U}	upper bound per scenario; plain Monte Carlo with zero failures needs $N = 3/p^{*}$ runs, so importance sampling is required	runs, ESS
simulation credibility	real-versus-simulated agreement for the same θ; NATM includes a credibility assessment framework [8, web article]	proving ground, replays
R_{unknown}^{U}	field driving without unexplained HB, Eq. (3) restricted to what the taxonomy misses	hours

Scenario-based testing lacks a direct quantitative link to mileage and an agreed coverage threshold [10, Sec. V, RA "Scenario Based Testing", Table XI]. It shrinks the field-testing target but does not remove it [10, Sec. VII].

4.5 Parameterising logical scenarios

1. Define the scenario type and its parameters, following the logical/concrete distinction of ISO 34502 [2, location not verified].
2. Keep θ small (3–6 dimensions) by sensitivity analysis; Glasmacher et al. list parameter sensitivity analysis in simulation as data-driven evidence [16, Sec. III-D, Table I].
3. Mine instances from drive logs with a scenario tagger [17, Secs. III-A–III-C]; each detection gives one θ sample and one count towards ν.
4. Fit f_{ℓ}: marginals plus copula (Sec. 5) or kernel density estimation [14, Ch. 3 and Ch. 6].
5. Estimate $\widehat{\nu}_{\ell} = n_{\ell}/T$ with a Poisson or Gamma interval.
6. Check the partition: no instance may belong to two logical scenarios.

4.6 Where each conditional probability comes from

Table 3: Sources of the terms in Eq. (5).

Quantity	Primary source	Caveat
$\nu_\ell, f_\ell(\theta)$	own fleet logs; public naturalistic data only if the ODD matches	system-dependent, e.g. exposure depends on routing and ODD [9, Sec. IV-B(b)–(c)]
$P(\mathrm{HB} \mid \theta)$	simulation with a perception error model learned from labelled field data; runtime monitors flag near-failure situations [12, Sec. 6.2, pp. 16–17]	simulation fidelity is limited [10, Sec. V, RA "Simulation-based Testing"]
$P(\bar{C} \mid E)$	for L4: evasion by other road users, via reactive agent models, or 1	an assumption to justify
$P(S \mid \mathrm{HB}, \theta)$	simulated Δv and impact angle, then an injury-risk curve, e.g. an omni-directional injury model [23, web page]	carry the curve's confidence band

5 Gaussian copula for scenario parameters

5.1 Definition

By Sklar's theorem, every joint distribution function can be written as $F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d))$ with a copula C [32, Sec. 2.3]. The Gaussian copula is $C_R(u) = \Phi_R(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d))$. Each scenario parameter keeps its own marginal, and the correlation matrix R alone sets the dependence.

5.2 Sampling and density

Sampling: $\varepsilon \sim \mathcal{N}(0, I)$, $Z = L\varepsilon$ with $R = LL^\top$, $U_i = \Phi(Z_i)$, $X_i = F_i^{-1}(U_i)$. The copula density, needed for the weights in Eq. (6), is

$$c_R(u) = |R|^{-1/2} \exp\Big(-\tfrac{1}{2} z^\top (R^{-1} - I) z \Big), \quad z = \Phi^{-1}(u), \qquad f(x) = c_R(F(x)) \prod_i f_i(x_i). \tag{8}$$

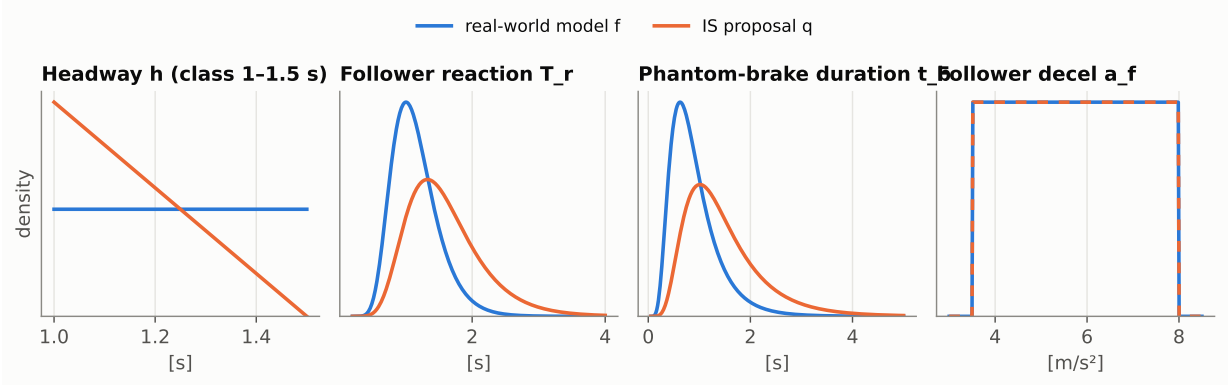


Figure 3: Marginals of the real-world model f and the importance-sampling proposal q used in Sec. 6[S].

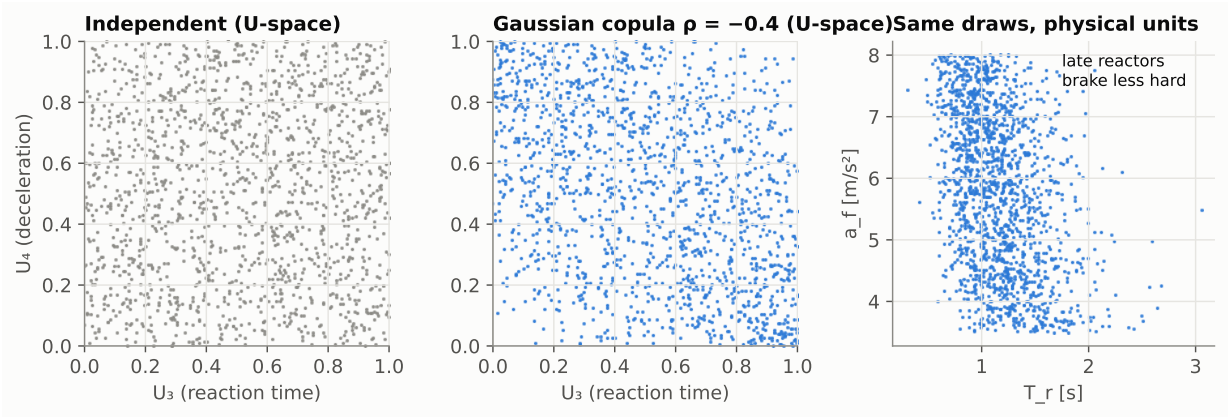


Figure 4: Independent uniforms, the Gaussian copula with $\rho = -0.4$, and the same draws in physical units[S].

5.3 Fitting

Transform each parameter to pseudo-observations $u = \text{rank}/(n+1)$ and normal scores $z = \Phi^{-1}(u)$, and estimate R as their correlation matrix. Pairwise, Kendall's τ [32, Ch. 5] converts to $\rho = \sin(\pi\tau/2)$ for elliptical copulas [29, location not verified]. R must be positive definite; pairwise estimates may need projection to the nearest correlation matrix.

5.4 Limitation: no tail dependence

For $\rho < 1$ the Gaussian copula has zero upper tail dependence, whereas the t-copula with ν degrees of freedom has $\lambda_U = 2\,t_{\nu+1}(-\sqrt{(\nu+1)(1-\rho)/(1+\rho)}) > 0$ [30, location not verified]. Joint extremes, such as a late reaction combined with weak braking, are therefore under-represented if real data show tail dependence (Fig. 5).

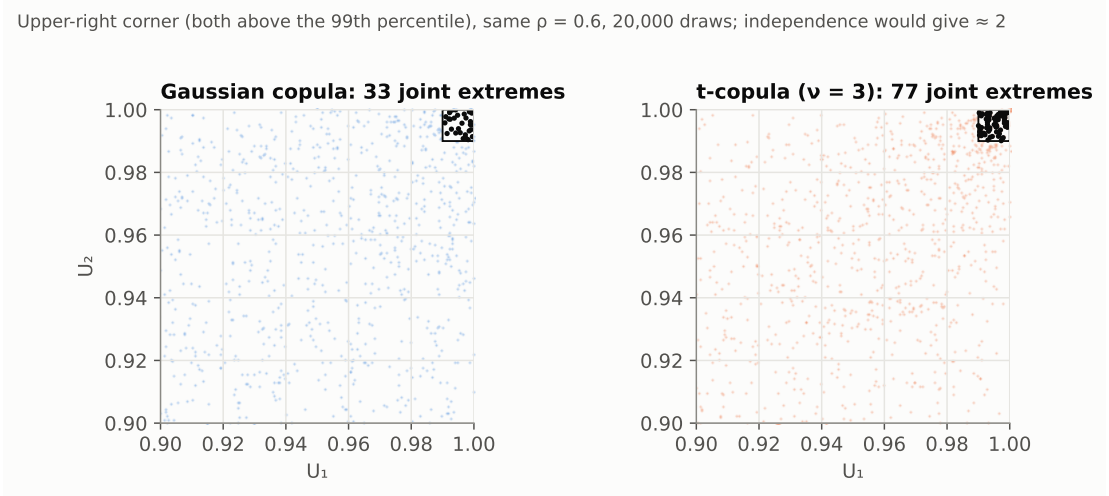


Figure 5: Same correlation, different joint extremes[S]: 33 (Gaussian) versus 77 (t, $\nu = 3$) draws in the top-1% corner out of 20,000; independence would give about 2 *(this report, computed)*.

6 Worked example: unintended braking with 1,000 concrete scenarios

6.1 Set-up

Everything in this section is synthetic[S]. Validation target: MAIS2+ injuries from rear-end collisions caused by unintended (phantom) braking $\leq A_H = 10^{-7}$/h, with 10% reserved for unknown triggers, leaving a budget of 9×10^{-8}/h. The ODD is partitioned into 100 logical scenarios: 5 trigger contexts × 4 ego speed bands × 5 follower-headway classes. The hazardous-behaviour rate λ depends on trigger and speed only, so 20 evidence cells serve 100 scenarios. Per logical scenario, 2,000 plain runs from f and 10 importance-sampled runs from q are simulated with a one-dimensional kinematic model ($\Delta t = 0.01$ s); ego speed, follower headway, reaction time and deceleration are coupled by a Gaussian copula with $\rho(v,h) = 0.25$, $\rho(h,T_r) = -0.30$, $\rho(T_r,a_f) = -0.40$. The illustrative injury curve is $P(\text{MAIS2+} \mid \Delta v) = 1/(1+e^{-(\Delta v-45)/7})$, Δv in km/h *(this report, App. A)*.

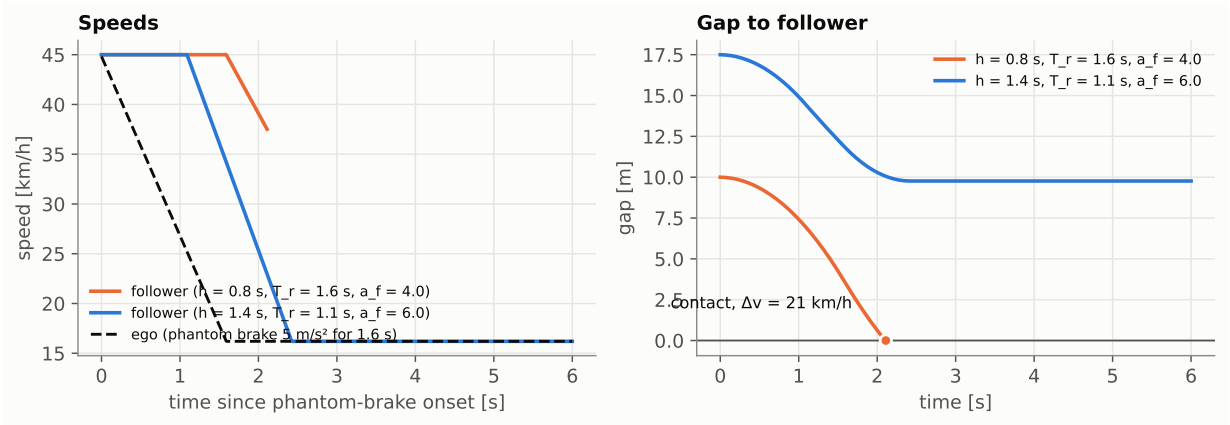


Figure 6: Two concrete scenarios with the same phantom brake[S]: contact with a short headway and a late, weak reaction; no contact otherwise.

6.2 Checks of the importance sampling

On 200,000 proposal draws the mean weight is 0.996, and the IS estimate (0.0161) matches 100,000 plain runs (0.0157). Over the 1,000 concrete scenarios the mean weight is 0.98, weights range from 5×10^{-4} to 67, 117 runs end in a collision, and the median ESS per logical scenario is 4.3 of 10 *(this report, Fig. 7)*.

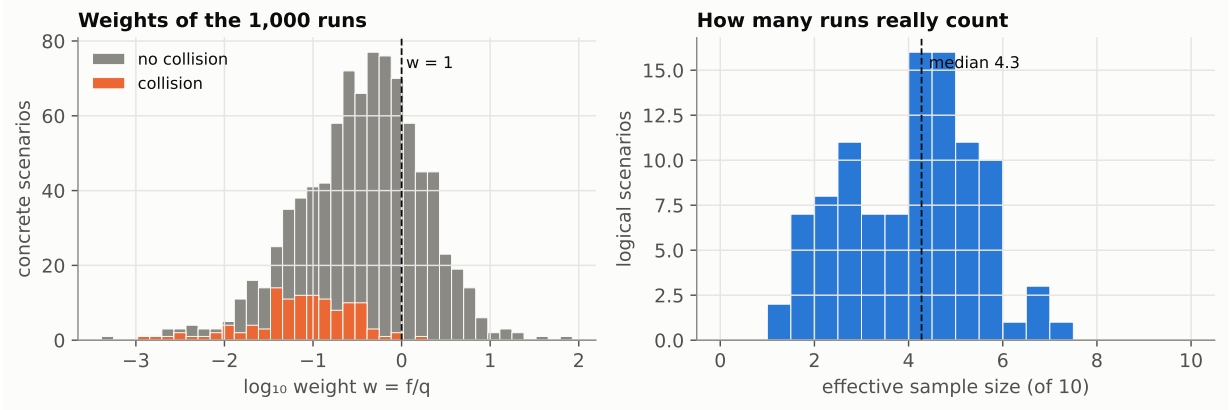


Figure 7: Weights of the 1,000 runs and effective sample size per logical scenario[S].

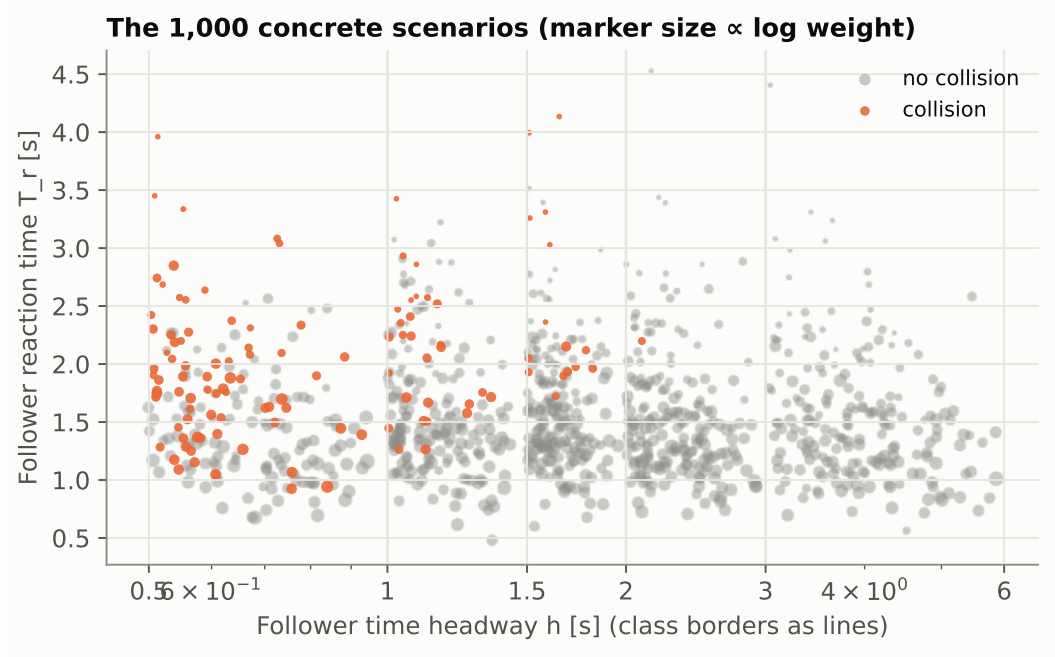


Figure 8: All 1,000 concrete scenarios[S]; collisions concentrate at short headway and late reaction.

6.3 Results

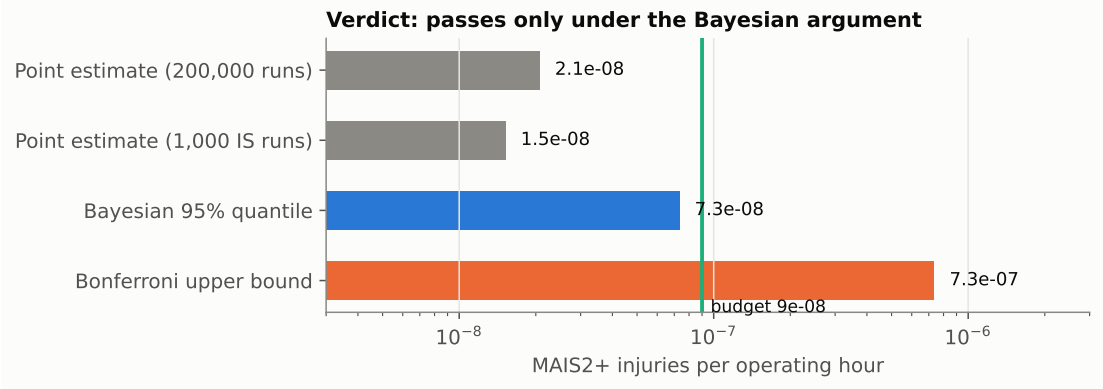


Figure 9: Verdict[S]: point estimates 2.1×10^{-8} (200,000 runs) and 1.5×10^{-8}/h (1,000 IS runs), Bayesian 95% quantile 7.3×10^{-8}/h (passes), Bonferroni bound over 120 estimates 7.3×10^{-7}/h (fails) *(this report, computed)*.

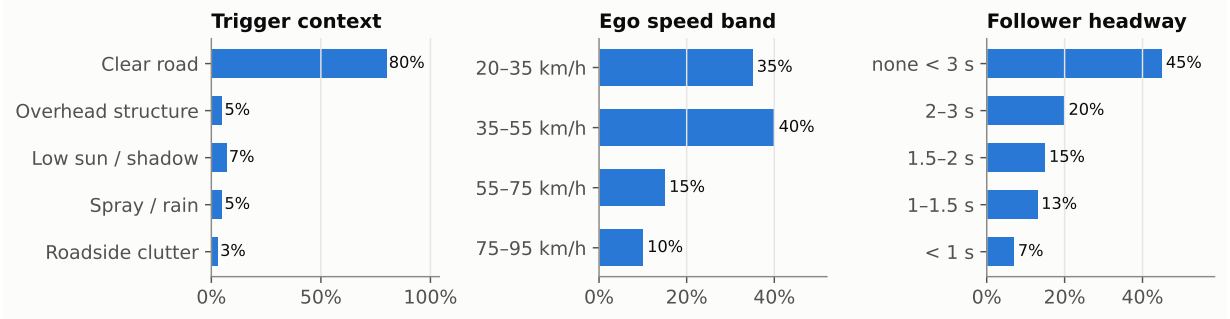


Figure 10: Exposure shares per factor[S]; scenario share = product (independence assumed).

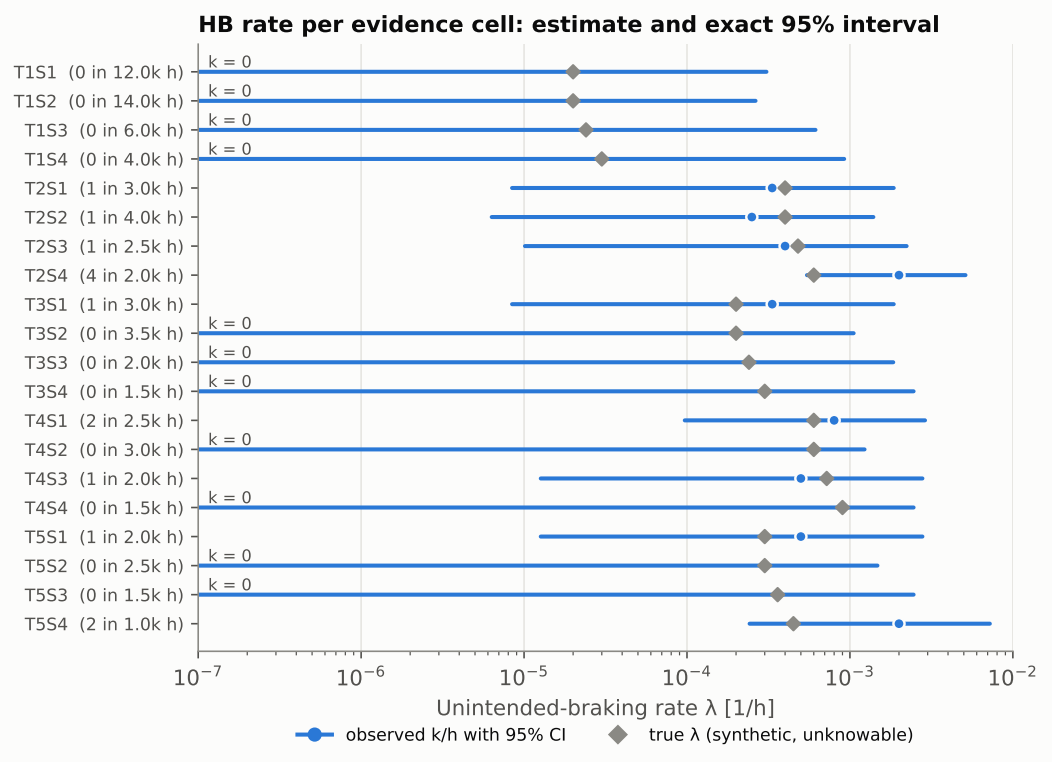


Figure 11: Hazardous-behaviour rate per evidence cell[S]: observed k/h with exact 95% interval, and the synthetic true rate.

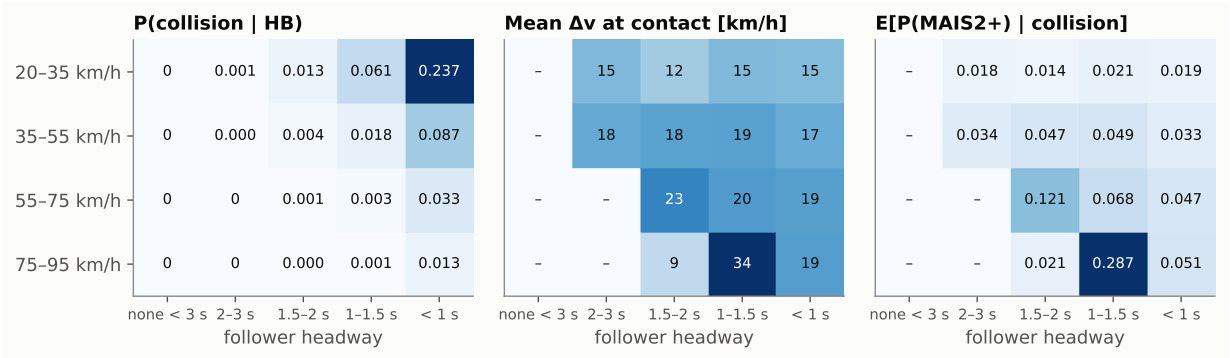


Figure 12: Collision probability, mean Δv and injury probability by speed band and headway[S].

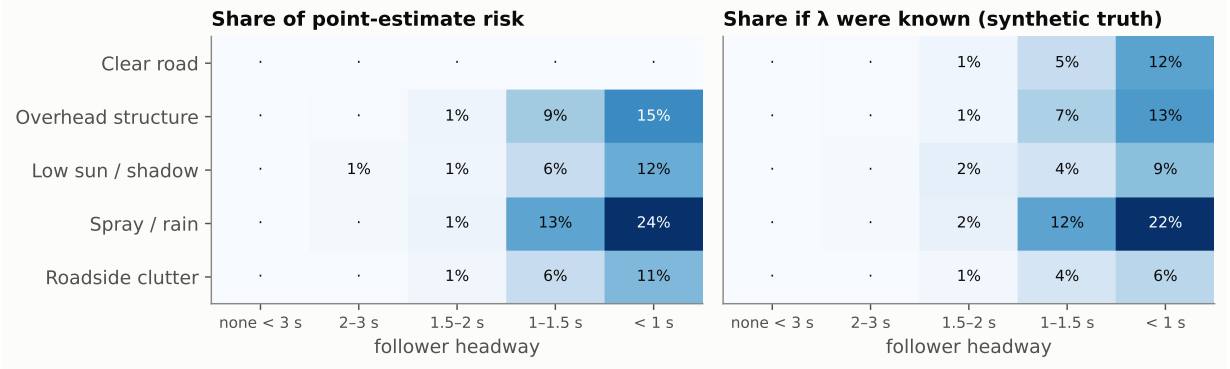


Figure 13: Where the risk sits[S]: attribution from observed counts (left) against attribution with the true rates (right). With 0–4 events per cell, point-estimate attribution is largely Poisson noise.

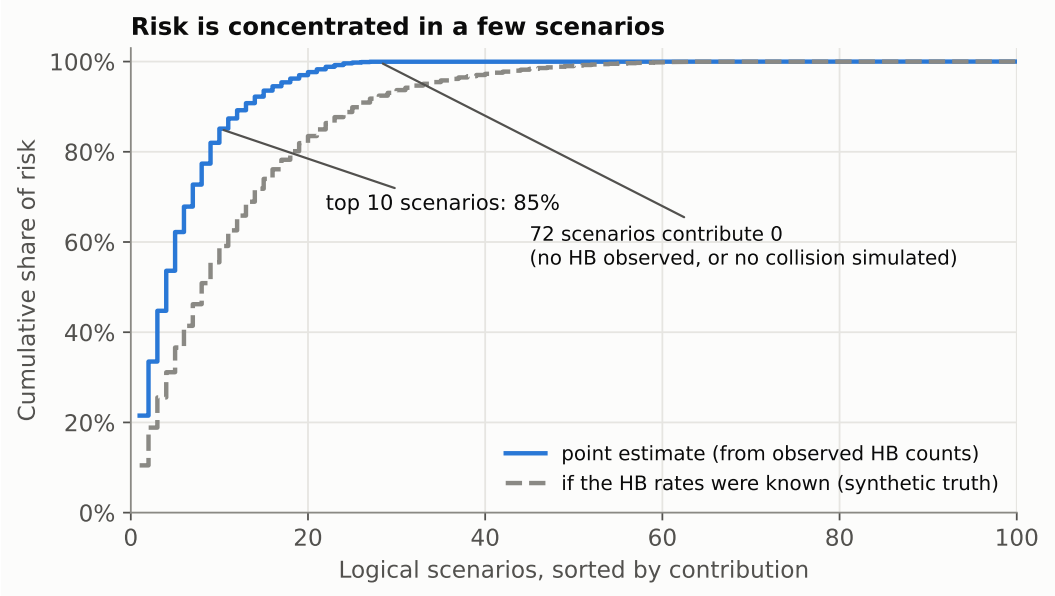


Figure 14: Cumulative share of the point estimate over the 100 logical scenarios[S]; the top 10 carry 85%.

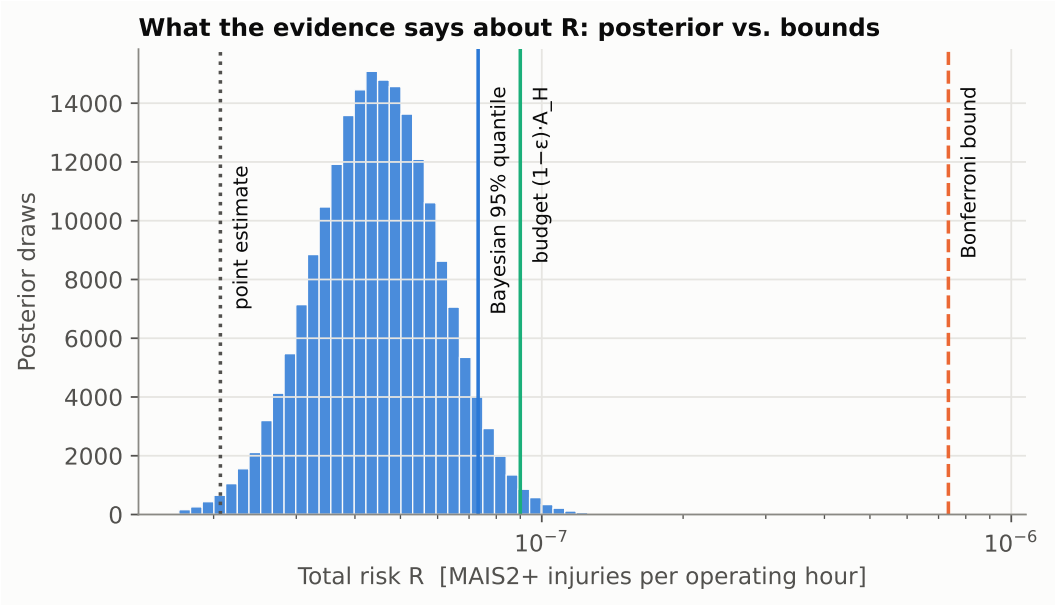


Figure 15: Posterior of total risk with Jeffreys priors[S], against point estimate, budget and Bonferroni bound. Jeffreys and flat priors as discussed in [10, Sec. V, RA "Bayesian Statistics"].

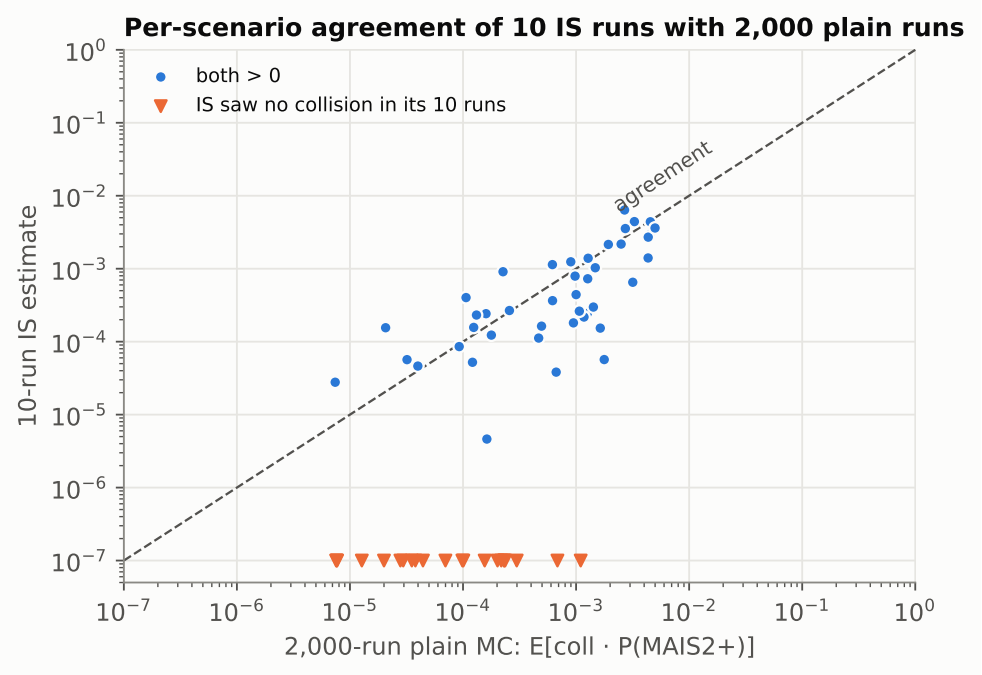


Figure 16: Per-scenario agreement of 10 IS runs with 2,000 plain runs[S]. 22 scenarios saw no collision in their 10 IS runs although plain Monte Carlo did.

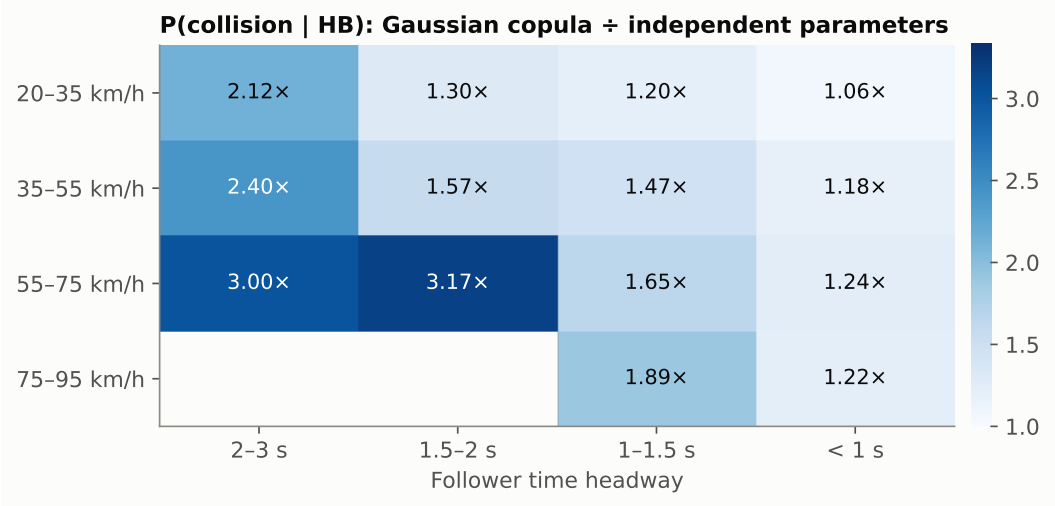


Figure 17: Ratio of $P(\text{collision} \mid \text{HB})$ with the Gaussian copula to the same model with independent parameters, 20,000 runs per cell[S]: 1.06 to 3.2.

6.4 Evidence needed

Minimising total test hours $\sum_j \lambda_k G_j / B_j$ subject to $\sum_j B_j = B$, with G_j the harm weight of cell j, gives $B_j \propto \sqrt{G_j}$ and $T_{\text{total}} = \lambda_k (\sum_j \sqrt{G_j})^2 / B$ *(this report, Lagrange multiplier)*. For the Bonferroni claim, 567,800 h are required against 73,500 h available (Fig. 18).

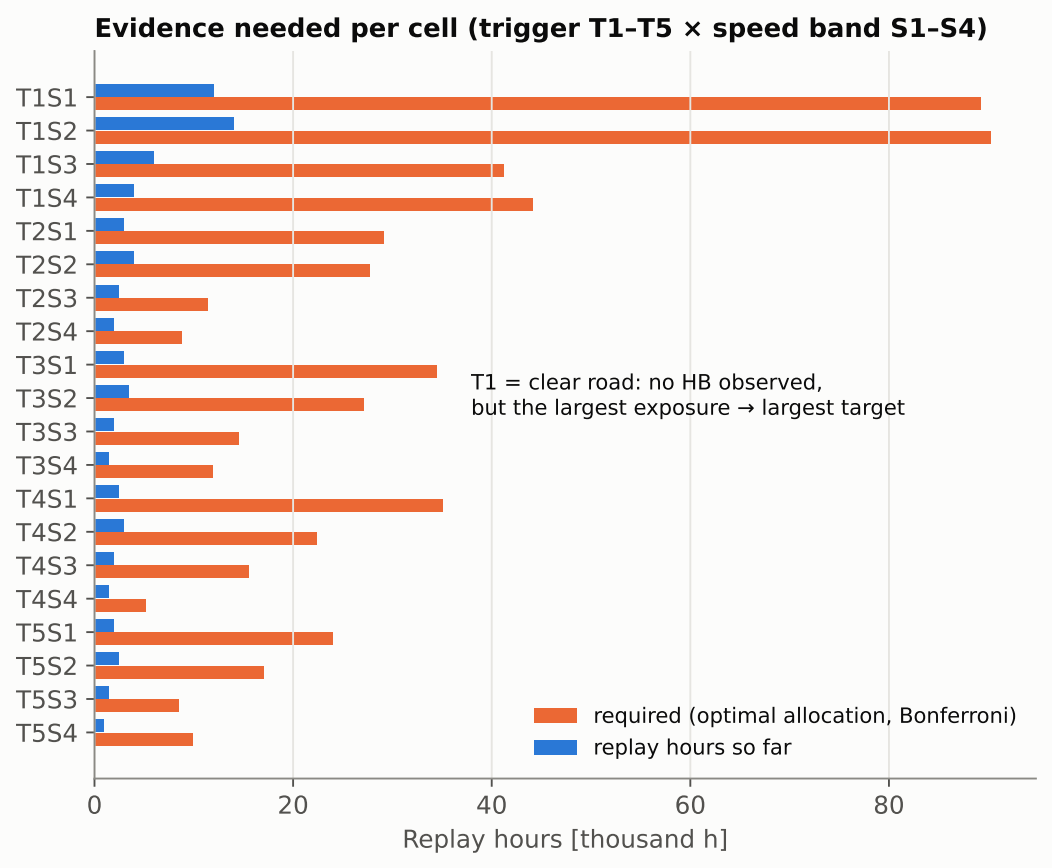


Figure 18: Replay hours per evidence cell, available versus required under the square-root allocation[S].

7 Field data, statistics and safety performance indicators

7.1 Data to collect

Table 4: Field data organised by the term of Eq. (5) it feeds.

Feeds	Data	How
exposure	hours and km with ODD tags per time step, including periods without any scenario	continuous signals plus map and weather lookup
scenario distributions	every occurrence with its parameters	scenario tagger [17, Sec. III]
HB rate	every brake request with its cause, plus suppressed near-HBs	event-triggered raw-data buffer
perception error model	misses and false positives against labelled ground truth, by class, range, occlusion, weather	labelled samples
outcomes	collisions with Δv, near-miss surrogates (min TTC, PET)	rear sensors, incident process
responses	minimal-risk manoeuvres, remote assistance, monitor flags [12, Sec. 4]	system logs
provenance	software hash, configuration, map version, calibration	attached to every hour

Clause 13 lists, as examples of monitored information, incidents that caused harm, exceedances of safety-relevant thresholds, and observations that extend the knowledge of scenarios and conditions [12, Sec. 2.1, p. 10].

7.2 Statistics

Rates are bounded with Eq. (4) per stratum and software version (Fig. 19); binomial proportions with Clopper–Pearson bounds [25, location not verified]. Near misses can be extrapolated to crash frequencies with extreme value analysis: a generalised Pareto distribution is fitted to exceedances over a threshold [33, Ch. 4]; Betschinske et al. review its use for validation and its open issues [10, Sec. V, RA "Extreme Value Analysis", Table IX].

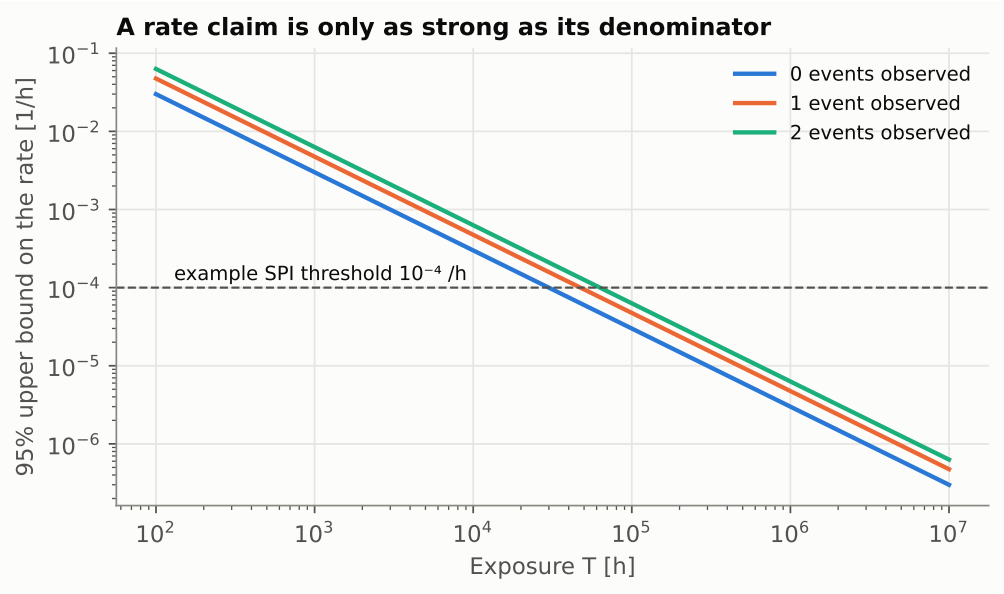


Figure 19: 95% upper bound on a rate against exposure for 0, 1, 2 observed events *(this report, Eq. (4))*.

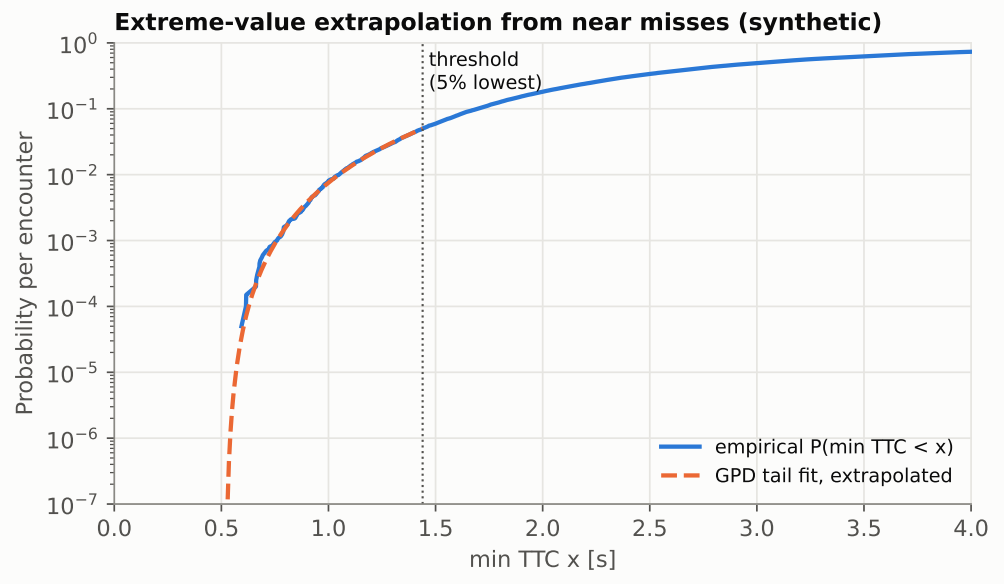


Figure 20: Extreme-value extrapolation of synthetic minimum-TTC data[S]: the lowest 5% are fitted with a generalised Pareto tail and extrapolated.

7.3 Safety performance indicators

UL 4600 defines: "An SPI is a metric supported by evidence that uses a threshold comparison to condition a claim in a safety case" (UL 4600, as cited in [21, Sec. 2, "UL4600"]), and states that "relying solely on lagging indicators is not acceptable" (UL 4600 note 16.2.2.6.2, as cited in [21, Sec. 2]). Lagging indicators detect shortcomings from past events; leading indicators identify potential issues before losses, e.g. unexpected fail-safe activations [21, Sec. 2]. Koopman contrasts KPIs ("how well do we drive") with SPIs ("how often are we potentially unsafe") [22, slide 12], lists lagging SPIs [22, slide 6] and leading SPIs [22, slide 7], and gives examples such as a false negative for more than k consecutive frames and violations of a safe clearance limit [22, slide 10]. An SPI violation means its claim is violated, but a claim can be violated without its SPI firing [21, Sec. 2; Sec. 3.1]. Responses range from updating the safety case and SPIs, through restricting operation (reduced speed, geofences) and updating the system, to grounding the fleet [21, Sec. 3.3].

Table 5: Example SPI register for unintended braking *(this report, Sec. 6)*.

Type	SPI	Claim conditioned	Threshold
lagging	rear-end collisions with ego struck, per h, by severity	overall target	A_H
leading	unexplained brake requests per 1,000 h per trigger context	HB rate per context	λ target, Fig. 18
leading	suppressed false-positive brake detections	margin before HB	trend alarm
leading	rear TTC < 1.5 s within 3 s after ego braking	$P(\text{coll} \mid \text{HB})$	simulation prediction
safety case	share of time with follower < 1 s	exposure assumption	χ^2 test
safety case	Kendall's τ(headway, reaction)	copula assumption	CI contains assumed ρ
safety case	rate of unclassifiable scenarios	reserve ε	Eq. (11)

The time an SPI needs to detect a doubling of its rate depends only on the expected count: for a one-sided exact Poisson test at $\alpha = 0.05$ and power 0.8 it takes $\mu^* \approx 9.1$ expected events, so $T = \mu^*/\lambda$ *(this report, computed)*. Lagging SPIs near the acceptance criterion would take decades to centuries for a 50-vehicle fleet (Fig. 21).

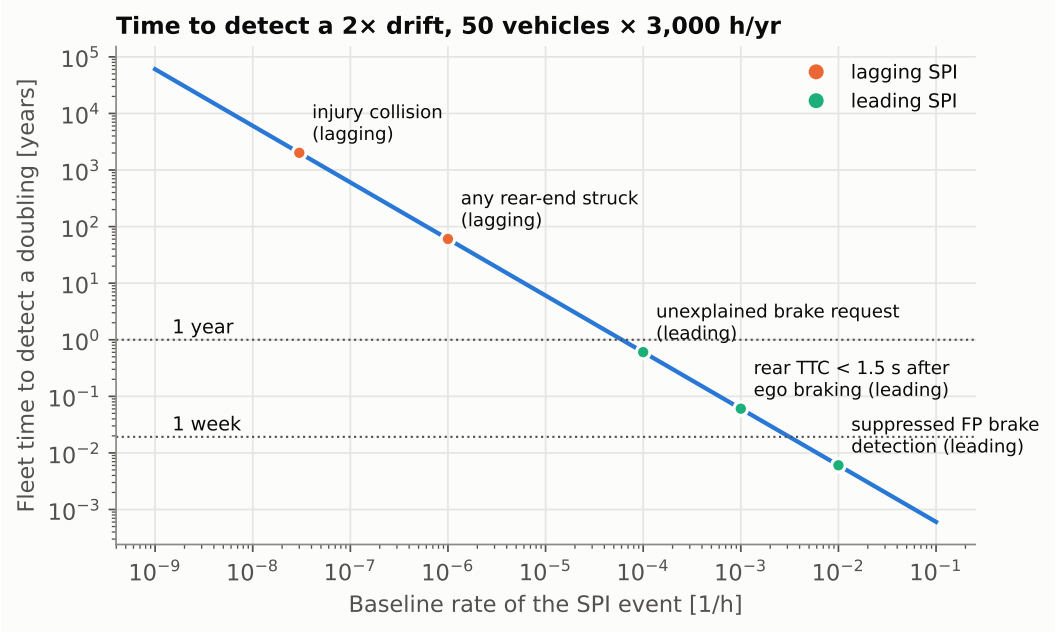


Figure 21: Fleet time to detect a 2× drift for 50 vehicles × 3,000 h/yr *(this report, computed).*

8 L4 validation, the scenario database and completeness

8.1 Vehicle-level SOTIF strategy

Table 6: Pillars of the vehicle-level strategy and the term they feed.

Pillar	SOTIF	Feeds
ODD spec, HARA, triggering-condition analysis	Cl. 5–7	A_s, HB/HE list, scenario taxonomy
component V&V: perception KPIs per ODD stratum, planner checks	Cl. 10	perception error models
scenario-based simulation from the database	Cl. 10	p_ℓ, coverage [12, Sec. 6.2, pp. 16–17]
log replay, XiL, proving ground	Cl. 10	λ per context; sim-to-real bounds
public-road exploration and FOT	Cl. 11	ν, f, R_{unknown} [12, Sec. 6.3, p. 17]
shadow mode, operation	Cl. 13	SPIs [12, Sec. 7, p. 18]
release argument	Cl. 12	Eq. (7) per harm

The structure matches the NATM pillars of virtual, track and real-world testing, audit and in-service monitoring [8, web article]. Schaller et al. show how a runtime monitor ensemble contributes evidence to Clauses 6, 9–11 and 13 without by itself establishing SOTIF compliance [12, Sec. 1.2, p. 3; Secs. 5–7, pp. 14–18], and propose that the standard make explicit that residual risk can be quantified from a statistical argument over traffic situations sourced from monitoring data [12, Sec. 8.2, p. 19].

8.2 What the scenario database contributes

1. It makes "known" explicit: the taxonomy is the boundary between known and unknown scenarios.
2. It holds the weights ν_ℓ and $f_\ell(\theta)$, derived from recordings.

3. It generates tests and keeps traceability from triggering condition to hazardous behaviour, hazardous event, requirement and test.
4. It shows coverage per scenario × triggering condition × software version.
5. It stores every run with θ, $q(\theta)$, software hash and outcome, so risk can be recomputed after changes.
6. It moves unknown to known: discovered types become categories.

```
Recording(id, vehicle, sw_hash, map_ver, calib_state, route, hours, raw_uri)
ExposureSlice(recording_id, t0, t1, odd_tags...) -- denominators
ScenarioCategory(id, version, tag_rule, param_schema, partition_key)
Occurrence(id, recording_id, t0, t1, category_id, theta..., odd_tags, min_ttc, hb_flag)
HazardLink(category_id, hb_id, he_id, triggering_condition_id)
TestRun(id, category_id, theta..., q_density, platform, sw_hash, outcome, dv)
```

8.3 From recordings to the database

Raw logs are turned into a reference world model by offline perception and labels, enriched with map, weather and ODD tags for every time step, tagged with actor activities and interactions, and converted into scenario occurrences with parameters read at defined anchor events. This is the tag-based pipeline of Guo et al.: preprocessing [17, Sec. III-A], tagging of actor activities [17, Sec. III-B.1] and interactions [17, Secs. III-B.2, III-B.3], and categorisation by tag combinations [17, Sec. III-C], applied to the Waymo Open Motion Dataset and CARLA data [17, Sec. IV]. Four rules matter: extract with a reference world model, not the stack under test; tag every driven second; attach provenance and a pointer to the raw log window; version the taxonomy and re-tag historical data when it changes.

8.4 Identifying scenarios automatically

Known types. Guo et al. report no incorrect extraction among 116 randomly examined of 215,090 extracted WOMD scenarios, but no precision or recall of individual tags [17, Sec. IV]. A tagger is a classifier, and its errors bias exposure. With measured precision and recall, $\hat{\nu}_\ell = n_{\text{detected}} \cdot \text{precision}/(\text{recall} \cdot T)$ *(this report, Fig. 22)*.

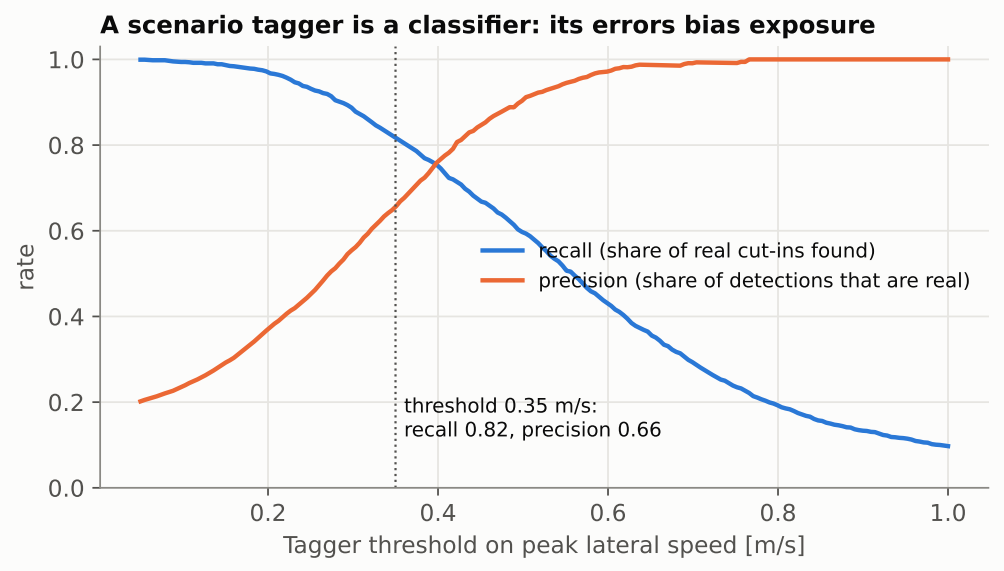


Figure 22: Synthetic cut-in tagger[S]: at 0.35 m/s it finds 82% of real cut-ins, and 66% of its detections are real *(this report, computed)*.

New types (Clause 11). (1) A queue of unexplained critical events (low TTC, hard braking, HB flags, monitor flags, disengagements) that match no category, for human review; Schaller et al. propose monitors that flag novel input/output behaviour and unfamiliar activation patterns as candidates for hazard identification [12, Sec. 5.1, pp. 14–15]. (2) Unsupervised clustering of time-window embeddings with outlier flags [18, Abstract]. (3) ODD novelty: conditions outside the specification or rare tag combinations.

8.5 ODD tags: coverage and granularity

ISO 34503 specifies the ODD taxonomy [3, location not verified] and ISO 34504 a tag-based scenario categorisation; ISO 34504 "provides examples rather than instructions on its application and tags" and does not fix values such as the speed difference in "moving faster or slower" [5, web article]. Both are a shared vocabulary, not a sufficient resolution. Tags missing for a given function (overhead structures, road spray or low sun for phantom braking) are derived from the triggering-condition analysis and from structured environment models: Reich's SEM use guide phrases with an environment ontology to identify the environment factors that influence hazardous situations and their risks [13, Sec. 1.3, p. 4], giving a hazard-specific environment model with explicit coverage checks [13, Sec. 1.4, contribution 1.1, p. 6; Sec. 4.1, p. 49].

Granularity criterion. A class may lump values together if a shift of the distribution inside the class cannot move the risk materially:

$$|\Delta R_c| = \nu_c \left| \int_c p(\theta)(f_c^{\text{dep}} - f_c^{\text{test}})d\theta \right| \leq \nu_c \, (p_c^{\max} - p_c^{\min}) \, \mathrm{TV}(f_c^{\text{dep}}, f_c^{\text{test}}) \tag{9}$$

(this report, derived). Classes are refined until the sum over classes fits a reserved share δ of the budget. In the example, $P(\text{collision} \mid \text{HB})$ falls from 0.137 to 0.030 inside the shortest headway class, while the 2–3 s class is flat (Fig. 23).

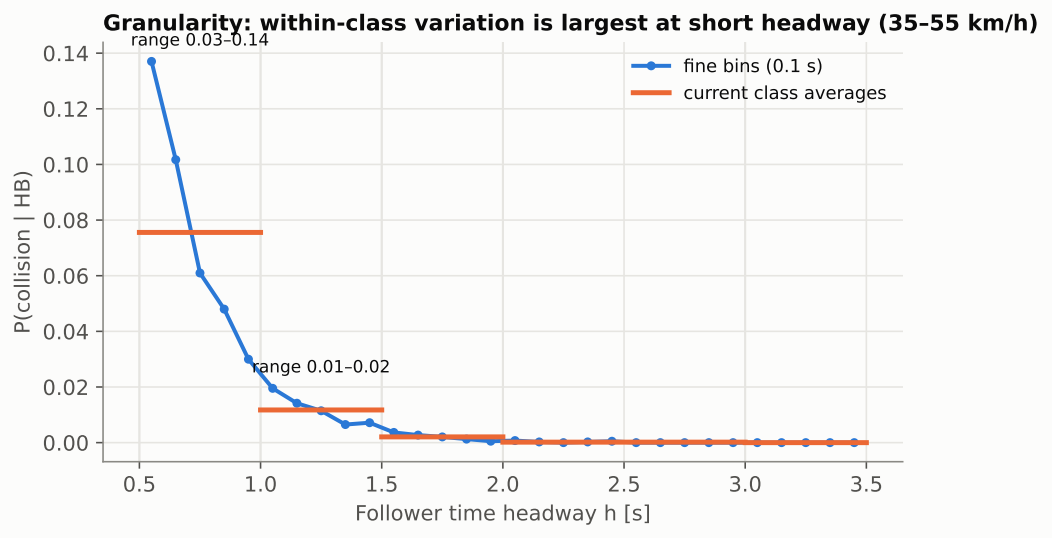


Figure 23: $P(\text{collision} \mid \text{HB})$ on a 0.1 s grid against current headway classes[S].

Combinatorial coverage. De Gelder et al. define tag-based coverage

$$\text{Coverage}_{\text{tag}}(n) = \frac{1}{n|L||C|} \sum_L \sum_C \min\left(n, N(L, C)\right) \tag{10}$$

[15, Sec. III-A, Eq. (1)], together with time-based, actor-based and actor-over-time coverage [15, Secs. III-B–III-D, Eqs. (2)–(4)]. On HighD with 10 categories and 18 tags, $\text{Coverage}_{\text{tag}}(10) = 1$

[15, Sec. IV-B], while about 75% of time instants are covered by at least one scenario [15, Sec. IV-C]. Applied to tag pairs, the metric exposes untested combinations (Fig. 24).

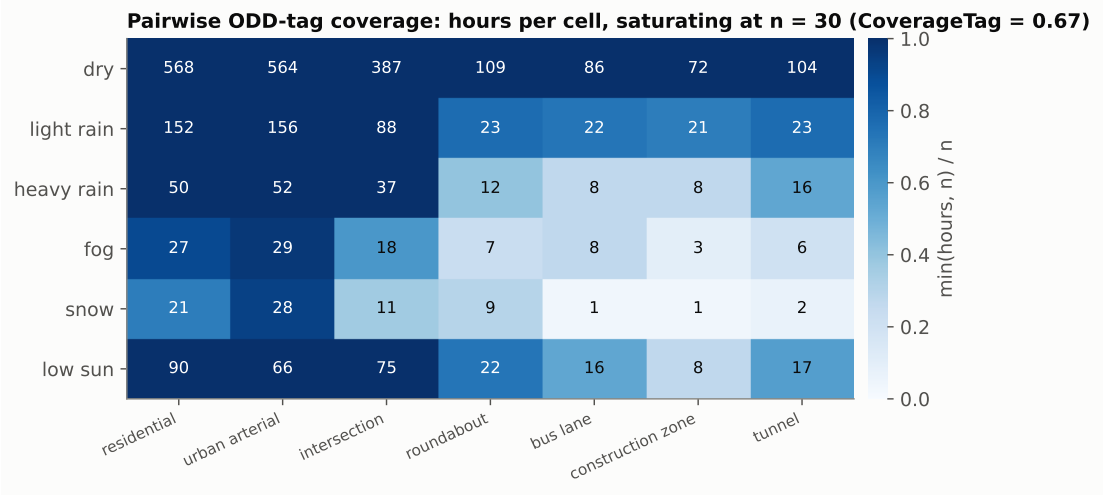


Figure 24: Synthetic exposure hours per weather × road-type pair[S]: 26 of 42 pairs have fewer than 30 h; $\text{Coverage}_{\text{tag}}(30) = 0.67$ *(this report, computed).*

8.6 Arguing completeness

Glasmacher et al. argue the completeness of a scenario concept in goal structuring notation by testing counter-hypotheses [16, Sec. III-C; applied in Sec. IV-B]. They define completeness as binary and applying to the concept, and coverage as quantifiable and applying to parameter spaces and value assignments [16, Secs. III-A, III-B]; evidence is knowledge-based or data-driven [16, Sec. III-D, Table I]. On inD they assign 59,253 base scenarios with no unassigned seconds and use saturation curves [16, Sec. IV-C, Fig. 5], noting that "unforeseen scenarios cannot be identified with argumentation, but with external input" [16, Sec. IV-C] and that completeness claims can expire and need review in a dev-ops process [16, Sec. V]. De Gelder et al. distinguish Type I coverage (all relevant aspects of the ODD) from Type II (all relevant aspects in the driving data) [15, Sec. III, Table III] and leave completeness metrics to future work [15, Sec. V]. Hauer et al. [19] estimate the number of scenario types not yet addressed in testing (as summarised in [15, Sec. II]).

Five legs of an argument follow:

1. construction: systematic derivation of the taxonomy, independent review;
2. explained time: the share of driven time assigned to a category (Type II coverage);
3. saturation: a flattening discovery curve and small missing mass;
4. risk-weighted remainder: bounded by Clause 11 field evidence within the reserve εA_H;
5. monitoring in operation, because completeness claims expire.

For leg 3, the Good–Turing estimate of the probability that the next occurrence is of an unseen type [26, location not verified] and the Chao1 richness estimator [27, location not verified] are

$$\hat{M}_0 = \frac{f_1}{N}, \qquad \hat{S} = S_{\text{obs}} + \frac{f_1(f_1-1)}{2(f_2+1)}, \qquad R_{\text{unknown}} \lesssim \hat{M}_0\, \nu_{\text{tot}}\, \bar{p}_{\text{unk}} \tag{11}$$

with f_1, f_2 the numbers of types seen once or twice and N the number of occurrences.

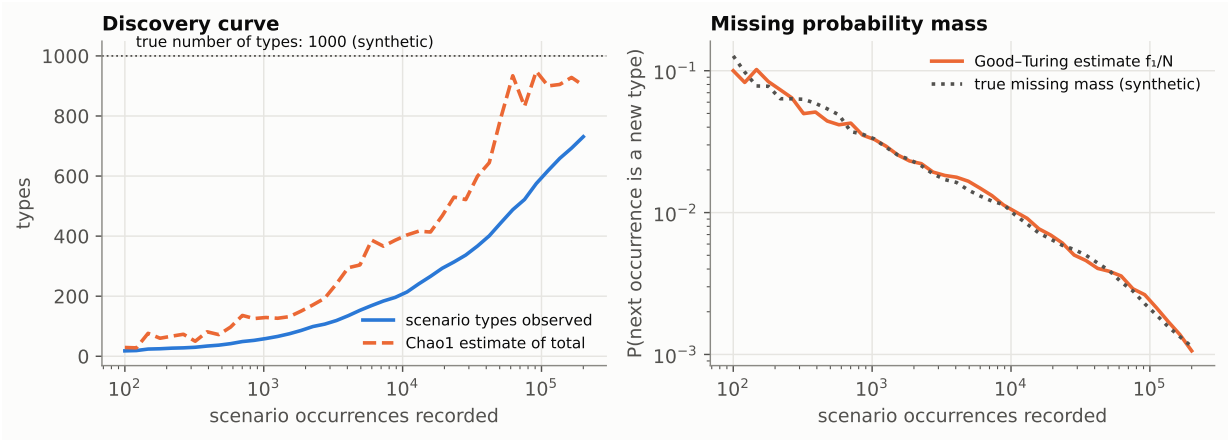


Figure 25: Synthetic: 1,000 scenario types with Zipf frequencies (exponent 1.8)[S]. After 200,000 occurrences, 731 types are seen and Chao1 estimates 902, a lower bound; the Good–Turing missing mass is 1.06×10^{-3} (true 1.14×10^{-3}) *(this report, computed).*

8.7 OpenLABEL and ISO 34504

Table 7: ASAM OpenLABEL compared with ISO 34504.

	ASAM OpenLABEL 1.0.0 (2021)	ISO 34504:2024
nature	data format (JSON schema) for annotations [6, standard page]	vocabulary and structure for scenario categories [5, web article]
content	elements (objects, actions, events, contexts, relations) with name, uid, type and ontology id [7, Sec. 2.3.1]; relations as RDF triples [7, Sec. 2.3.4]; frames and frame intervals [7, Sec. 2.3.2]; streams [7, Sec. 2.3.6]; coordinate systems [7, Secs. 2.3.7, 3.1]	tags for vehicle types, environments, light, weather, road conditions, grouped into categories; examples, no exact values [5, web article]
semantics	none of its own: elements point to external ontologies [7, Sec. 2.3.3]	a semantic source to point to
scenario tags	labels selected from a pool and stored with scenario files for search [7, Secs. 5.1–5.2]	tags define the categories
related	lists BSI PAS 1883 and ASAM OpenODD [7, Sec. 1.3]	part of the ISO 3450x family [5, web article]

They complement each other: ISO 34503/34504 plus custom tags form the ontology; OpenLABEL stores per-recording, per-interval annotations that reference it; the scenario database indexes them; OpenSCENARIO describes executable tests.

9 Scaling to ACC and an L4 robotaxi

The estimator (5) does not change; the scenario space, the agent interaction and the cost of exploring rare regions do.

Table 8: Three levels compared *(this report, assessment)*.

	unintended braking (Sec. 6)	ACC / L2 longitudinal	L4 urban robotaxi
hazardous behaviours	1	about 5–10	dozens across all DDT subtasks
scenario types	1	follow, lead braking, cut-in, cut-out, queue approach	hundreds: intersections, occluded VRUs, pick-up/drop-off, emergency vehicles
parameters	7	about 10–20	about 50–100 and more
interaction	open loop	closed loop, driver as fallback	closed loop, many reactive agents
acceptance	one A_H	per HE, human benchmark (Annex C)	positive risk balance per harm type (ISO/TS 5083)

ACC is specified in ISO 15622 [4, location not verified]. With d parameters shifted by δ standard deviations, a Gaussian proposal gives $E_q[w^2] = e^{d\delta^2}$ and $\mathrm{ESS}/N \approx e^{-d\delta^2}$ *(this report, derived; Fig. 26)*. Fixed-proposal importance sampling therefore degrades in high dimension, which motivates the cross-entropy method [34, location not verified], subset simulation [28, location not verified] and adaptive importance sampling [31, Ch. 10]. Runtime risk models can restrict exposure to risky situations: SINADRA derives monitorable risk-implausibility hypotheses from HARA work products and turns them into Boolean or Bayesian-network runtime models [13, Sec. 1.3, pp. 4–5; Sec. 4.3, p. 77]; in its urban case study, behaviour-intent likelihoods weight a collision-risk signal [13, Sec. 5.2.5, p. 131]. Such a reduction of ν_ℓ must be argued inside Eq. (5).

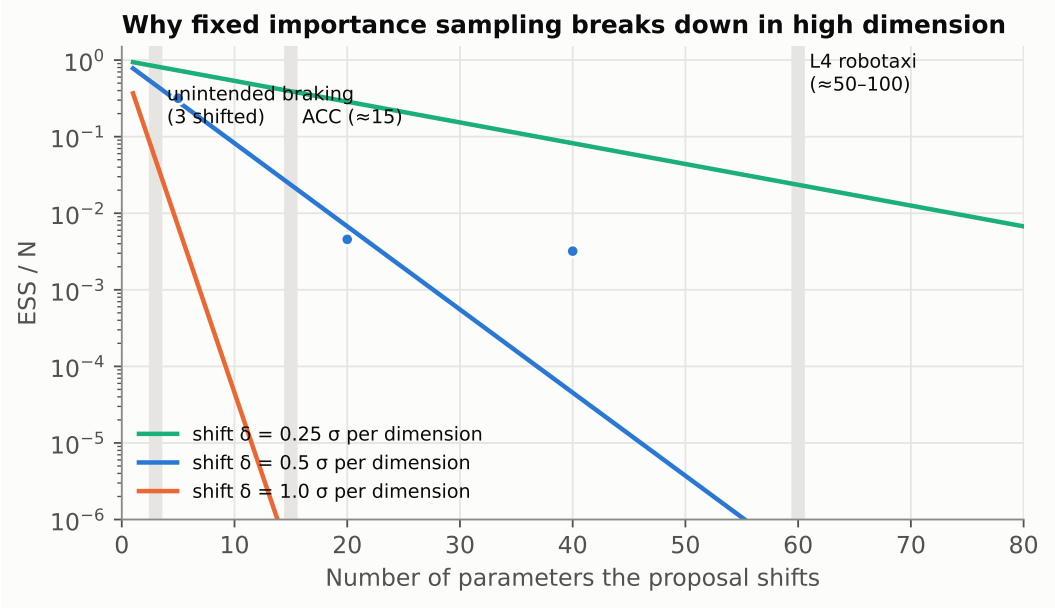


Figure 26: Effective sample size of importance sampling against the number of shifted parameters *(this report, derived and simulated)*.

10 Limitations

- All example data are synthetic; correlations, rates, distributions and the injury curve are illustrative.
- ISO clause and page references were not checked against the standards themselves (Sec. 1.2).

- The kinematic model is one-dimensional, without lateral escape, brake build-up or reaction to brake release.
- Scenario shares are products of factor shares (independence between trigger, speed and headway).
- The Bayesian result depends on the prior, which must be defended to an assessor.

A Code and reproduction

All numbers marked *this report* and all figures are reproducible from the repository `dfeldmann-bot/sotif-risk-course` (private): `tools/ub_sim.py` (scenario generation, copula, simulation, weights), `tools/build_ub_example.py` (workbook with live formulas), `tools/build_docs_html.py` (figures and HTML docs) and `tools/gen_report_figs.py` (the figures of this report). Random seed: 2026.

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- A. B. Owen (2013), see [31]: Ch. 9 for importance sampling diagnostics and effective sample size.
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